Project Report

1. INTRODUCTION

1.1 Project Overview

The project **Olympic Games Data Analysis** uses Tableau Public to analyse Olympic data and present meaningful insights through interactive visualisations, a dashboard, and a story. It examines medal records across athletes, sports, teams, genders, seasons, and host cities.

1.2 Purpose

The purpose of this project is to transform Olympic data into clear visual insights, identify trends and comparisons, and help users understand Olympic medal-record patterns.

2. IDEATION PHASE

2.1 Problem Statement

Olympic datasets contain information about athletes, sports, teams, events, seasons, and medal records. Analysing this information manually can make it difficult to identify patterns, compare performance, and understand trends. This project addresses the problem by creating visualisations that make Olympic data easier to explore and interpret.

2.2 Empathy Map Canvas

- **Says:** Users want to understand Olympic medal records and compare performances.
- **Thinks:** Users want to know which teams and sports have higher medal-record counts.
- **Does:** Users explore charts, compare categories, and examine trends.
- **Feels:** Users want data to be clear, accessible, and easy to understand.

2.3 Brainstorming

The project uses different visualisation techniques to analyse Olympic data. The planned visuals include comparisons by sport, gender, season, team, and host city, along with athlete-level analysis and trends over time. These visuals are combined into a dashboard and a Tableau Story.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. The user opens the Olympic Games Data Analysis dashboard.
2. The user reviews the overall medal-record summary.
3. The user explores individual charts to compare teams, sports, genders, and seasons.
4. The user examines medal-record trends and host-city comparisons.
5. The user uses the Tableau Story to follow the findings in a structured sequence.

3.2 Solution Requirement

- An Olympic dataset containing athlete, sport, team, year, season, city, gender, and medal-record information.

- Tableau Public Desktop for data analysis and visualisation.
- Nine visualisations addressing different questions about the dataset.
- A dashboard combining the visualisations.
- A Tableau Story explaining the analysis and findings.
- Tableau Public publishing access to share the final project.

3.3 Data Flow Diagram

Olympic Dataset → Data Connection → Data Inspection and Preparation → Tableau Worksheets → Nine Visualisations → Interactive Dashboard → Tableau Story → Published Project

3.4 Technology Stack

- **Data Source:** Olympic dataset
- **Data Visualisation:** Tableau Public Desktop
- **Data Analysis:** Tableau dimensions, measures, aggregations, sorting, and filters
- **Dashboard Development:** Tableau Dashboard
- **Data Storytelling:** Tableau Story
- **Publishing:** Tableau Public
- **Documentation and Submission:** PDF templates, GitHub, and SkillWallet

4. PROJECT DESIGN

4.1 Problem Solution Fit

The project provides a visual solution for exploring Olympic medal-record data. Instead of manually examining rows of data, users can compare categories and identify patterns through charts, a dashboard, and a story. This makes the data easier to interpret and present.

4.2 Proposed Solution

The proposed solution is an Olympic Games Data Analysis dashboard developed using Tableau Public. It includes nine visualisations covering average medal records per athlete, top teams, trends over time, gender comparisons, seasonal comparisons, sport-wise records, gender and season analysis, and host-city comparisons. The Tableau Story presents these visualisations in a logical sequence.

4.3 Solution Architecture

1. **Data Source:** Olympic dataset containing athlete and event information.
2. **Data Connection:** Connect the dataset to Tableau Public Desktop.
3. **Data Preparation:** Inspect available fields, data types, dimensions, and measures.
4. **Data Analysis:** Apply aggregations, calculations, sorting, and filters as required.
5. **Visualisation Layer:** Create nine worksheets to answer the selected analytical questions.
6. **Dashboard Layer:** Combine the worksheets into one dashboard.
7. **Story Layer:** Arrange the dashboard and individual visualisations into a sequence.
8. **Publishing Layer:** Publish the dashboard and story on Tableau Public.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

1. **Phase 1 – Data Understanding:** Identify the dataset fields and understand the available Olympic information.
2. **Phase 2 – Visualisation Planning:** Select the nine analytical questions and appropriate chart types.
3. **Phase 3 – Worksheet Development:** Create the nine visualisations in Tableau.
4. **Phase 4 – Dashboard Design:** Arrange the charts into a clear and readable dashboard.
5. **Phase 5 – Story Development:** Organise the visualisations into a narrative sequence.
6. **Phase 6 – Testing:** Check chart titles, labels, values, sorting, and dashboard readability.
7. **Phase 7 – Publishing:** Publish the dashboard and story on Tableau Public.
8. **Phase 8 – Documentation:** Prepare the project report, screenshots, and demonstration video.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The project was checked to ensure that:

- The Olympic dataset connects successfully to Tableau.
- The required dimensions and measures are available for analysis.
- All nine visualisations display correctly.
- Chart titles, labels, and axes are readable.
- Sorting, filters, and calculations work as intended.
- The dashboard presents the visualisations in an organised layout.
- The Tableau Story displays the selected worksheets in the intended order.
- The published dashboard and story links open successfully.

7. RESULTS

7.1 Output Screenshots

Insert screenshots of the following outputs:

1. Olympic Games Data Analysis dashboard.
2. Average Medal Records per Athlete by Sport.
3. Top 10 Olympic Teams by Medal Records.
4. Olympic Medal-Record Trends Over Time.
5. Olympic Medal Records by Gender.
6. Olympic Medal Records by Season.
7. Olympic Medal Records by Sport.
8. Olympic Medal Records by Gender and Season.
9. Top 10 Olympic Cities by Medal Records.
10. Tableau Story showing the visualisations in sequence.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Makes Olympic data easier to understand through visualisations.
- Enables comparisons across sports, teams, genders, seasons, and cities.

- Presents multiple analytical views in one dashboard.
- Uses a Tableau Story to explain findings in sequence.
- Can be shared online through Tableau Public.

Disadvantages

- The analysis depends on the completeness and quality of the dataset.
- The findings are limited to the fields and records available in the dataset.
- Medal-record counts should not automatically be interpreted as unique medal counts or an athlete's probability of winning.
- Tableau Public publishing requires an internet connection.
- The dashboard does not establish causal relationships between the variables.

9. CONCLUSION

The Olympic Games Data Analysis project demonstrates how Tableau Public can be used to explore and visualise Olympic data. The nine visualisations provide different perspectives on medal records across athletes, teams, sports, genders, seasons, years, and host cities. The dashboard brings the analysis together, while the Tableau Story presents the visual findings in a structured sequence. The project improves the accessibility and interpretability of the dataset through visual data analysis.

10. FUTURE SCOPE

- Add more Olympic editions and updated datasets.
- Improve interactivity with additional filters and drill-downs.
- Analyse country-wise performance across Olympic editions.
- Compare Summer and Winter Olympics over longer periods.
- Add more detailed athlete-level and sport-level analysis.
- Include additional performance indicators where the dataset supports them.
- Improve dashboard layout and mobile readability.

11. APPENDIX

Source Code: Tableau workbook containing the worksheets, dashboard, and story.

Dataset Link: Insert the original dataset URL used for this project.

GitHub & Project Demo Links:

- **GitHub Repository:**
  https://github.com/bhavariyavaishnavi-hub/global-olympic-sports-analysis
- **Tableau Public Dashboard:**
  https://public.tableau.com/app/profile/vaishnavi.bhavariya/viz/Global_Olympic_Sports_Analysis_twb2/Dashboard1
- **Tableau Public Story:**
  https://drive.google.com/file/d/1VMqJ8Dbnuw6-EpKFkyn0-e8K3WiOAWdh/view?usp=drive_link